

EVERETT D. MONTGOMERY

Principal Software Architect & Director of Java Engineering

San Francisco, CA • (415) 555-8291 • e.montgomery.architect@enterprise-java.io
linkedin.com/in/everett-montgomery-java-lead • US Citizen

PROFESSIONAL SUMMARY

Technically formidable Principal Software Architect and Engineering Leader with over 12 years of experience in designing, developing, and deploying enterprise-scale distributed systems using the Java ecosystem. Distinguished for architecting the "Lumina-Cloud" core engine, which supports over 500 million concurrent WebSocket connections with sub-20ms latency. Everett is a visionary leader who has successfully managed engineering organizations of 120+ developers, bridging the gap between low-level JVM optimization and high-level strategic business objectives. Expert in transitioning monolithic legacy architectures into cloud-native microservices environments, consistently achieving a 40% reduction in infrastructure costs and a 600% increase in deployment frequency.

Recognized as a thought leader in reactive programming (Project Loom, Vert.x) and high-performance computing within the Java space. Proven track record of managing annual R&D budgets exceeding \$160M while directing cross-functional teams across North America, EMEA, and APAC. Thaddeus possesses deep expertise in Spring Boot 3.x, Quarkus, Kubernetes, and event-driven architectures utilizing Kafka and Pulsar. A champion of "Clean Architecture" and TDD, Everett has successfully navigated the complexities of SOC2 and PCI-DSS compliance for Tier-1 financial institutions. His leadership philosophy centers on technical mentorship, radical transparency, and the pursuit of architectural excellence through rigorous peer review and automated quality gates.

CORE COMPETENCIES

- **Architectural Leadership:** Microservices, Event-Driven Architecture (EDA), Hexagonal/Clean Architecture, Domain-Driven Design (DDD).
- **Java Ecosystem Mastery:** JVM Tuning (G1GC, ZGC, Shenandoah), GraalVM, Spring Framework 6.x, Hibernate/JPA, Project Loom.
- **Distributed Systems:** Service Mesh (Istio), API Gateway (Kong, Spring Cloud Gateway), Distributed Caching (Hazelcast, Redis).
- **Engineering Leadership:** Agile/Scrum Transformation, Engineering OKRs, P&L Management (\$100M+), Talent Acquisition & Leveling.
- **DevOps & Cloud:** Kubernetes (K8s) Operator Pattern, Terraform (IaC), AWS (EKS, Lambda, Aurora), Google Cloud Platform (GCP).
- **Data Engineering:** Apache Kafka (KSQL, Streams), Cassandra Modeling, PostgreSQL Optimization, Elasticsearch/ELK.
- **Security & Compliance:** OAuth2/OIDC, JWT Security, mTLS, OWASP Top 10 Mitigation, HIPAA & PCI-DSS Frameworks.

- **Quality Engineering:** Test-Driven Development (TDD), Consumer-Driven Contract Testing (Pact), Chaos Engineering (Gremlin).

TECHNICAL SKILLS

Java & Backend Engineering Stack

- **Languages:** Java 17/21 (Expert), Kotlin, Scala, Go (Golang), Python, Bash, SQL, Groovy.
- **Frameworks:** Spring Boot 3.x (Security, Data, Cloud), Quarkus, Micronaut, Vert.x, Jakarta EE, Akka.
- **Reactive Stack:** Project Reactor, RxJava 3, Mutiny, Akka Streams.
- **Build & Dependency:** Maven, Gradle (Kotlin DSL), Nexus, Artifactory.

Distributed Data & Messaging

- **Relational Databases:** PostgreSQL (PostGIS, Citus), Oracle 19c, MySQL, CockroachDB (Distributed SQL).
- **NoSQL Platforms:** MongoDB, Apache Cassandra, Neo4j (Graph), DynamoDB, ScyllaDB.
- **Messaging/Streaming:** Apache Kafka (Expert), RabbitMQ, Amazon SQS/SNS, Apache Pulsar, Redis Pub/Sub.
- **Search Engines:** Elasticsearch 8.x, Solr, Meilisearch.

Cloud, DevOps & Infrastructure

- **Orchestration:** Kubernetes (Expert), OpenShift, Docker, Podman.
- **Service Mesh:** Istio, Linkerd, Consul Connect.
- **Infrastructure as Code:** Terraform, AWS CDK, Pulumi, Ansible, Packer.
- **CI/CD Platforms:** GitHub Actions, Jenkins (Groovy Pipelines), GitLab CI, Spinnaker.
- **Monitoring & Logging:** Prometheus, Grafana, Datadog, New Relic, Splunk, Loki.

Architectural Standards & Tools

- **Methodologies:** Agile, Kanban, Safe Framework, Scrum of Scrums, LeSS (Large Scale Scrum).
- **Tools:** IntelliJ IDEA Ultimate, JProfiler, YourKit, VisualVM, Wireshark, Postman, Swagger/OpenAPI.
- **Design Patterns:** SAGA, CQRS, Event Sourcing, Circuit Breaker, Bulkhead, Sidecar, Strangler Fig.

PROFESSIONAL EXPERIENCE

Principal Java Architect & Director of Engineering | Lumina Global Cloud | 2021 – Present

San Francisco, CA / Remote (Tier-1 Cloud Infrastructure Provider | Annual Revenue: \$18B)

Executive Leadership & Strategic Oversight: Appointed as the primary technical authority for the "Core-Services" division, managing a multi-regional engineering organization of 125 SDETs and Software Engineers. Accountable for the architectural integrity of the global identity and data-transit layers. Manage a \$165M annual budget covering headcount, cloud infrastructure, and R&D investments.

Key Measurable Achievements:

- **Global WebSocket Transformation:** Architected a reactive Java-based gateway using Vert.x and Netty to handle 500M+ persistent connections. Reduced memory footprint per connection by 70%, saving the company \$45M in annual EC2 costs.
- **Zero-Trust Identity Migration:** Led the transition from legacy session-based auth to a modern OIDC/OAuth2 framework utilizing mTLS and SPIFFE/SPIRE. Achieved 100% SOC2 Type II compliance across all internal microservices.
- **JVM Performance Engineering:** Implemented a standardized ZGC (Z Garbage Collector) tuning profile across 4,000+ microservice instances. Reduced p99 latency from 150ms to 8ms for critical path services.
- **Cloud-Native Modernization:** Spearheaded the migration from a monolithic Oracle-backed application to a distributed CockroachDB and Spring Boot 3 architecture. Increased system availability from 99.9% to 99.995% ("Five Nines" target).
- **Engineering Productivity:** Introduced an internal "Backstage.io" portal with automated Java project scaffolding. Reduced the "Idea-to-Production" time for new microservices from 3 weeks to 15 minutes.
- **Talent Development:** Established the "Lumina Architecture Guild," a formal mentorship program that upskilled 25 Senior Engineers to Lead/Architect levels, resulting in a 92% engineering retention rate.
- **Financial Optimization:** Leveraged GraalVM Native Image compilation for Lambda functions, reducing cold start times by 95% and reducing serverless spend by \$4.2M annually.

Director of Java Development | Titan Financial Tech | 2017 – 2021

New York, NY (High-Frequency Trading & Digital Banking Infrastructure)

Operational Leadership: Directed the engineering efforts for the "Titan-Trade" platform, an institutional-grade digital asset exchange. Led a team of 80 backend engineers focused on low-latency order matching and risk management systems.

Key Measurable Achievements:

- **Matching Engine Architecture:** Designed an in-memory Java matching engine utilizing LMAX Disruptor and off-heap memory management. Achieved a throughput of 2 million orders per second with a deterministic latency profile.
- **Event Sourcing Implementation:** Re-architected the clearing and settlement system using Event Sourcing (Axon Framework) and Kafka. Provided 100% auditability for every financial transaction, satisfying rigorous SEC regulatory requirements.
- **Microservices Reliability:** Implemented a comprehensive resilience strategy using Resilience4j (Circuit Breakers, Rate Limiters). Successfully maintained 100% uptime during the 2020 market volatility spike where volumes reached 15x normal levels.
- **Big Data Interoperability:** Built a real-time risk analytics engine using Spark Streaming and Java 11. Processed \$2B in daily volume with real-time "Value at Risk" (VaR) calculations updated every 500ms.
- **CI/CD Transformation:** Orchestrated the shift from manual Jenkins jobs to fully automated Groovy-based pipelines with integrated SonarQube and Checkstyle gates. Reduced production defect leakage by 45%.
- **Regulatory Compliance:** Led the technical team through 4 successful PCI-DSS Level 1 audits. Implemented automated data masking and field-level encryption for all PII data in Cassandra.

Senior Software Architect (Lead) | SilverPath Healthcare | 2014 – 2017

Chicago, IL (Health Information Exchange & Interoperability Solutions)

Technical Leadership: Served as the Lead Architect for the "SilverPath Interop" platform, connecting 450+ hospitals and clinics. Focused on HL7/FHIR data standards and large-scale data ingestion.

Key Measurable Achievements:

- **FHIR API Infrastructure:** Developed a high-performance HAPI-FHIR based server in Java, enabling real-time clinical data exchange for 12M+ patients. Recognized as an industry benchmark for FHIR implementation.
- **ETL Pipeline Optimization:** Redesigned the data ingestion pipeline using Spring Batch and Apache Camel. Reduced the daily ingestion window from 12 hours to 2 hours while handling 5x the data volume (fictional client: *Great Lakes Health Network*).
- **Security Hardening:** Implemented a specialized HIPAA-compliant audit logging system using Elasticsearch. Enabled forensic-level tracing of all PHI access with sub-second query performance.
- **Cloud Migration Lead:** Directed the "Lift-and-Shift" of the legacy Java EE application stack to AWS (Elastic Beanstalk), reducing physical data center costs by \$8M over 3 years.

Senior Java Developer | Swift-Logistics Analytics | 2011 – 2014

Dallas, TX (Supply Chain Optimization SaaS)

Key Measurable Achievements:

- **Monolith Decomposition:** Acted as the lead developer for extracting the "Route-Optimization" engine into a standalone Spring Boot microservice. Improved engine performance by 300% through multi-threaded optimization.
- **Database Refactoring:** Led the migration of 500 million records from a legacy SQL Server instance to a sharded PostgreSQL cluster, ensuring zero downtime during the cutover.
- **Automated Testing Advocate:** Introduced JUnit 5 and Mockito to the engineering team. Increased code coverage from 15% to 85% across the core logistics modules.

KEY PROJECTS PORTFOLIO

Project Lumina-Core: Reactive Data Fabric

Role: Chief Architect | **Duration:** 24 Months

Objective: Build a multi-tenant, distributed data plane capable of handling exabyte-scale telemetry for global cloud clients.

- **Strategy:** Utilized a "Shared-Nothing" architecture with Java 17 and Netty. Implemented a custom consensus algorithm based on Raft for distributed state management.
- **Implementation:** Developed a reactive client-side SDK that optimized TCP connection pooling and payload compression. Used Java's 'MemorySegment' API for efficient zero-copy data transfer.
- **Outcome:** Enabled a throughput of 100GB/s per cluster. Cited in a fictional *Gartner* report as the leading "Cloud Data Ingress" architecture of 2023.

Project Titan-Match: Low-Latency Exchange

Role: Principal Architect | **Duration:** 18 Months

Objective: Replace a legacy C++ matching engine with a high-performance Java implementation to improve developer velocity.

- **Strategy:** Leveraged "Mechanical Sympathy" principles. Optimized Java code for CPU cache line alignment and minimized object allocation to avoid GC pauses.
- **Implementation:** Built a custom Aeron-based transport layer for low-latency UDP communication between the matching engine and risk checks.
- **Outcome:** Achieved a sub-50 microsecond mean latency profile. Proved that Java can match C++ performance for financial applications when architected with memory awareness.

Project Green-Build: Automated Developer Ecosystem

Role: Engineering Director | **Duration:** 12 Months

Objective: Standardize the Java development lifecycle for 500+ engineers globally.

- **Implementation:** Built a custom Maven plugin that automatically injects security vulnerability scanning (Snyk) and licensing checks into the build cycle.
- **Outcome:** Reduced critical security vulnerabilities in production by 92%. Saved an estimated 5,000 developer hours per year in manual compliance checks.

Operation Cloud-Bridge: Monolith to Serverless

Role: Strategy Lead | **Outcome:** Saved \$12M in annual infrastructure spend.

- **Strategy:** Used the "Strangler Fig" pattern to migrate legacy monolithic components to Java-based AWS Lambda functions.
- **Action:** Optimized cold starts using GraalVM and Provisioned Concurrency models.

LEADERSHIP & TEAM MANAGEMENT

- **Scaling Engineering Organizations:** Successfully grew the "Core-Services" team at Lumina from 20 to 125 engineers in under 2 years. Developed a distributed hiring model with technical hubs in Austin, Berlin, and Bangalore.
- **The "Vance" Quality Framework:** Authored a 150-page internal standard for Java development, covering everything from 'Optional' usage to distributed tracing patterns. This framework is now used by 1,200+ developers company-wide.
- **OKR-Driven Growth:** Implemented a "Value-Stream" focused OKR framework that aligned engineering sprints with direct commercial revenue targets. Resulted in a 35% increase in feature delivery velocity.
- **Crisis Leadership:** Managed the technical response to a major industry-wide zero-day vulnerability (fictional variant of Log4Shell). Successfully patched 4,000+ services within 6 hours with zero exploitation of client data.
- **Executive Stakeholder Management:** Provide bi-weekly "Architectural Health" briefings to the CEO and Board of Directors, focusing on risk mitigation, scalability projections, and R&D ROI.
- **Diversity & Inclusion in Engineering:** Launched a "Pathways to Architect" program for underrepresented groups in tech, increasing female representation in the architecture guild by 50% in 3 years.

PROCESS IMPROVEMENTS & AUTOMATION

- **Automated JVM Profiling:** Built a "Continuous Profiling" tool using Java Flight Recorder (JFR) that automatically triggers a profile dump when CPU spikes occur in production. Reduced MTTR for performance issues by 80%.
- **Self-Healing Kubernetes Operators:** Developed a custom K8s Operator in Java that monitors microservice health and automatically adjusts JVM heap sizes and GC settings based on real-time traffic patterns.
- **Chaos Engineering Pipeline:** Integrated "Gremlin" chaos experiments into the staging environment. Automated the failure of individual Kafka brokers and network partitions, resulting in a 50% reduction in production S1 incidents.
- **Shadow Traffic Testing:** Developed a "Mirror-Execution" proxy that clones production traffic and sends it to a "Dark Launch" version of the Java microservice. Allowed for the verification of performance and logic without impacting real users.
- **Automated Compliance Evidence:** Built a tool that automatically exports security logs and dependency manifests into audit-ready PDF reports, reducing annual audit preparation time by 1,000 man-hours.

STRATEGIC VISION: THE FUTURE OF JAVA

The Era of Cloud-Native Java

I believe the next decade of enterprise software will be defined by the transition from "Java-on-the-Cloud" to "Cloud-Native Java." This involves moving away from massive, long-running JVMs to lightweight, GraalVM-compiled native binaries that can scale horizontally in milliseconds. My mission is to build the architectural primitives that make this transition seamless for global enterprises, ensuring that Java remains the premier language for the next generation of cloud infrastructure.

Architectural Resiliency in a Bipolar World

In an era of increasing geopolitical volatility, technical resiliency must include data sovereignty and multi-cloud portability. I champion a "Cloud-Agnostic" Java architecture where the application logic is strictly separated from the infrastructure via the Sidecar and Service Mesh patterns. I move organizations away from "Vendor Lock-In" to "Operational Sovereignty."

Ethical Engineering and AI

As we integrate Generative AI into our development workflows (Copilot, Whisper), I am focused on the "Ethical SDLC." We must ensure that our automated coding tools do not introduce bias or insecure patterns into our mission-critical Java systems. My strategy involves a "Human-in-the-Loop" architecture for AI-generated code, verified by rigorous automated security gates.

BOARD ENGAGEMENT & INDUSTRY ADVISING

- **Audit & Risk Committee Reporting:** Provide quarterly briefings to the Board of Directors at Lumina Global on system scalability, cyber-resilience, and the ROI of the \$160M R&D budget.
- **Industry Advisor:** Currently serve as an advisory board member for three early-stage Fintech startups, providing feedback on distributed database selection and cloud-native architecture.
- **Technical Due Diligence:** Acted as the lead technical evaluator for 6 successful acquisitions (totaling \$2.5B in deal value), assessing the target company's architectural health and technical debt.
- **Open Source Contributor:** Active contributor to the Spring Framework and Project Loom; frequently provide feedback to the Oracle Java Platform Group on future JVM enhancements.

EDUCATION

Master of Science in Distributed Systems & Software Engineering

Massachusetts Institute of Technology (MIT), Cambridge, MA

- Research Focus: Consistency Protocols in Large-Scale Distributed Databases.
- Thesis: "Dynamic Scaling Limits of Java Microservices in Non-Deterministic Environments."
- Honors: MIT Presidential Fellowship.

Bachelor of Science in Computer Science & Engineering

Stanford University, Palo Alto, CA

- Honors: Summa Cum Laude; President of the Stanford Computer Science Society.
- Recipient of the Terman Engineering Scholastic Award.

CERTIFICATIONS

- **AWS Certified Solutions Architect – Professional**
- **Oracle Certified Professional:** Java SE 17 Developer
- **CKA:** Certified Kubernetes Administrator (Cloud Native Computing Foundation)
- **Certified Information Systems Security Professional (CISSP)**
- **Google Cloud Professional Cloud Architect**
- **Spring Certified Professional** (Broadcom/VMware)
- **Apache Cassandra Certified Professional**

TOOLS & TECHNOLOGIES REFERENCE

Core Languages: Java 8/11/17/21, Kotlin, Go, Python.

Frameworks: Spring Boot, Quarkus, Micronaut, Vert.x, Akka, Camel, Batch.

Persistence: PostgreSQL, CockroachDB, Cassandra, MongoDB, Redis, Elasticsearch.

Streaming: Kafka (Streams, Connect), Pulsar, RabbitMQ, SQS/SNS.

DevOps: Kubernetes, Istio, Docker, Terraform, Helm, ArgoCD, GitHub Actions.

Observability: Prometheus, Grafana, Datadog, New Relic, Splunk, Honeycomb.

Testing: JUnit 5, Mockito, Testcontainers, Pact, RestAssured, k6, Gremlin.

PUBLICATIONS & CONFERENCES

Selected Speaking Engagements

- **Keynote Speaker:** "Java in the Age of Kubernetes: Architecting for High Density," *Devoxx US* 2023.
- **Featured Speaker:** "Project Loom: Transforming Scalability in Financial Systems," *JavaOne* 2022.
- **Panelist:** "Microservices vs. Modulith: The Great Architectural Debate," *QCon San Francisco* 2021.
- **Workshop Lead:** "JVM Performance Tuning for Low Latency," *SpringOne* 2020.
- **Guest Lecturer:** "Advanced Distributed Systems," *UC Berkeley - College of Engineering*.

Selected Peer-Reviewed Publications

- **"Predictive Garbage Collection: Utilizing Machine Learning for JVM Optimization"**
IEEE Journal of Software Architecture, 2023.
- **"Beyond REST: Comparing gRPC and RSocket for High-Frequency Java Services"**
The Journal of High Performance Systems, 2022.
- **"Architecting for Reliability: A Case Study of Distributed Event Sourcing in Fintech"**
International Software Engineering Journal, 2021.
- **"The Cold Start Problem: Native Image Compilation for Serverless Java"**
Cloud Computing Digest, 2020.

LANGUAGES & GLOBAL EXPOSURE

- **English:** Native Proficiency.
- **German:** Professional Working Proficiency (C1 Level).
- **Technical Contexts:** Expert in navigating international technical cultures and labor laws for engineering recruitment in the EU, APAC, and North America.
- **Global Footprint:** Directly managed distributed teams in: USA, Canada, Germany, Poland, UK, India, and Singapore.